

Brand image and equity: the mediating role of brand equity drivers and moderating effects of product type and word of mouth

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Abstract In today's competitive landscape, organizations are progressively searching for valuable practices to create strong consumer-based brand equity. Managers must remain vigilant and receptive to factors that influence brand equity. Accordingly, this paper addresses this issue. Specifically, the paper empirically investigates the mediating roles of three drivers of brand equity in the brand image—consumer-based brand equity relationship and explores the moderating roles of product type and word of mouth (WOM) on the linkages between brand image and brand equity. In order to understand these relationships, a questionnaire survey was administered among brand consumers, yielding a total of 321 responses. Confirmatory factor analysis and Cronbach's alpha were utilized to estimate the validity and reliability of the measures. Structural equation modeling with maximum likelihood estimation was employed to test the hypotheses. The authors find that brand image positively affects brand attachment, brand attitude, and brand awareness. Brand attachment and attitude positively influence consumer-based brand equity. The findings show that brand attachment and brand attitude partially mediate the relationship between brand image and brand equity. Product type and WOM have a significant moderating effect on the relationships among brand image, drivers of brand equity, and brand equity. The authors discuss the limitations of the study and provide future research directions.

Keywords Brand image · Consumer-based brand equity · Drivers of brand equity · Word of mouth · Contingent effects

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1 Introduction

Building strong brand equity is a key issue in today's business world. Consumer-based brand equity could lead to robust distribution networks, establish product brand extensions, and reinforce price elasticity (Kay 2006). Accordingly, marketers are giving particular attention to the future outlook of brands, and are increasingly searching for effective methods to create strong brand equity to remain competitive or avoid becoming obsolete (Wilcox et al. 2008). To meet these challenges, marketers and researchers have intensified the role of brand equity (Faircloth et al. 2001), as consumers are generally committed to a limited number of brands (Thomson et al. 2005). Obviously, effective branding that results in favorable consumer evaluations is a critical element in brand success.

Keller (1993) suggested that there are two types of brand equity measurements: financial-based and consumer-based. Nonetheless, Keller's consumer-based brand equity model provides a framework for how consumers' responses help in building brand equity (Kakati and Choudhury 2013). A number of studies have focused on this topic (e.g., Faircloth et al. 2001; Jaiprakash 2008; Mourad et al. 2011). Notwithstanding the attention the topic has received, one of the main concerns for marketers is the complex issue and inadequate understanding of consumer-based brand equity, particularly with respect to predicting consumer behavior toward the brand from a contingency perspective (Hamzaoui-Essoussi et al. 2011; Wang et al. 2006).

Another critical dimension of branding that has received considerable attention is brand image (Keller 2001). Researchers have generally viewed brand image as an important factor that affects consumers' purchase decisions. A good brand image strategy distinguishes a company's brand from its competitors, resulting in favorable evaluations and associations in the consumers' mind (Kumaravel and Kandasamy 2012). Accordingly, brand image is one of the most significant factors that builds brand equity. Previous studies (e.g., Faircloth et al. 2001; Jaiprakash 2008; Kim et al. 2003) have shown that brand image has a direct effect on brand equity; however, they have not indicated how this relationship exists. The existing literature has not yet identified the intervening mechanism that describes the path of influence from the brand image context to brand equity. It thus remains uncertain whether brand equity drivers play a significant role in connecting brand image and brand equity. Consequently, drivers of brand equity could be one of the prevailing mechanisms through which brand image influences brand equity. Thus, it is important to identify potential brand equity drivers (e.g., brand attachment, brand attitude, brand awareness) that shape the relationship between brand image and brand equity. Accordingly, decision makers could pay more attention to these drivers in order to boost consumer-based brand equity.

Very few researchers have tried to systematically investigate the moderating effects between brand image and brand equity. Identifying moderating effects is necessary, as it will enable scholars to understand whether differential relationships arise according to identifiable portions of their sample (Bryman and Cramer 1999). According to Christodoulides and Chernatony (2010), the brand category is

a critical factor in brand equity measurement. In this line, Lovett et al. (2013) claimed that product type may act as a moderating variable, and suggested that future research explore such frameworks. To the best of our knowledge, despite efforts to assess the effect of brand image on brand equity, no research to date has considered how such a relationship is affected by differences in product type. We propose that brand image per se cannot adequately explain all variations in brand equity, but rather that it is contingent on product type. Accordingly, among different types of products, such as search goods and experience goods, the effect of brand image on consumer-based brand equity may differ. As an example, Adidas and Nike brands, which belong to the clothing and shoes categories, are considered search goods, while Head & Shoulders and Pantene brands, which belong to the haircare category, are treated as experience goods. The effect of brand image on consumer-based brand equity may thus vary among these brands. For instance, what if Adidas and Nike were to focus more on their brand image compared to Head & Shoulders and Pantene? Or what if Adidas and Nike attempted to strengthen their brand equity drivers to a greater extent compared to Head & Shoulders and Pantene? Additionally, marketers wish to better understand word of mouth (WOM) because traditional forms of communication (e.g., advertising) are tending to lose their effectiveness (Trusov et al. 2009). In other words, WOM has become a much more influential factor on consumer behavior compared to other forms of marketing communications, such as advertising and publicity (Alam and Yasin 2010). Although WOM has been acknowledged as the most effective marketing strategy (Misner 1999), the role of WOM and its impact on the relationship between brand image and consumer-based brand equity has yet to be investigated.

This study attempts to fill these gaps. It is designed to test a conceptual model that incorporates the relationships between brand image and consumer-based brand equity. Specifically, this study has two main purposes. First, it acknowledges that both brand image and drivers of brand equity are vital to building strong brand equity. Using a path model, the study examines how brand equity drivers mediate the relationship between brand image and brand equity. Second, this study investigates the moderating role of product type and WOM on the relationships between brand image, drivers of brand equity, and brand equity. This study adds to the existing literature by assessing the linkages between important brand constructs, and examines the moderating impact of product type and WOM on brand equity. Understanding such relationships can have important implications for branding strategies. In this study, the model and hypotheses are conceptualized using associative network memory theory.

This paper proceeds in the following manner. First, the theory of associative network memory is described. The constructs of brand image, brand equity, and drivers of brand equity are then defined. Next, the research hypotheses and framework are introduced. This is followed by the research methodology, analysis, and results. We conclude with implications and directions for future research.

2 Conceptual framework

2.1 Associative network memory theory

Associative network memory theory (ANMT) relies on how memory works (Srull and Wyer 1989). Indeed, the theory assumes that memory contains pieces of information called nodes (Smith 2004). A node plays the role of activator for other nodes, either through external information obtained from external sources or internal information retrieved from long-term memory (Keller 1993). Once an individual perceives information from internal or external sources, it is decoded into an abstract, mental language and is allocated to a node in the network knowledge structure (Kreuzbauer and Malter 2005).

According to Keller (1993), memory principles influence understanding and knowledge about a brand, and ultimately affect brand equity. Understanding the structure of brand knowledge is critical because this knowledge influences consumers' when they think of the specific brand. That said, ANMT suggests that semantic memory consists of a set of nodes that are connected to each other by associations (Keller 1993). According to this theory, semantic information is ranked in consumers' minds in a hierarchical model, wherein nodes are connected by association (Morrin 1999). Accordingly, the relationships between brand image, drivers of brand equity, and brand equity, including the moderating effects of WOM and product type, could be explained by ANMT because a necessary condition for such links is establishing a brand node in a consumer's memory.

2.2 Consumer-based brand equity and its drivers

According to Keller (1993), brand equity can be defined in terms of marketing effects that are attributed to a brand; for example, particular behavioral responses from the marketing of a product due to its name that would not occur if the same product did not have a brand name. Accordingly, Keller (1993, p. 8) defined consumer-based brand equity as "the differential effect of brand knowledge on consumer response to the marketing of the brand". Consistent with this description, Keller (1993) noted that a brand has a positive (negative) consumer-based brand community if consumers respond more (less) enthusiastically to the product, price, or promotion than they do to the same marketing mix activities when associated with a fictitiously named version of the product. Therefore, measurement and examination of brand equity are important because they reflect the health of the brand and indicate whether the firm's brand management is effective (Shabbir and Rehman 2013). Indeed, a brand with higher equity tends to acquire higher market share and prices can be positioned at a premium relative to competitors (Liao and Cheng 2014). Accordingly, the importance of brand equity has been highlighted by many researchers and marketers alike. For instance, Ailawadi et al. (2003), Broniarczyk and Alba (1994), and Loken and John (1993) have written extensively on the concept of brand equity and how it can be properly managed. Brand equity is measured using two approaches: financial-based and consumer-based (Keller 1993).

From a financial perspective, brand equity focuses on the value of a brand for the company. From the consumers' point of view, brand equity concentrates on the measurement of brand equity based on consumer behavior (Buil et al. 2013). Keller's consumer-based brand equity model provides a framework of how consumers' responses help in building brand equity (Kakati and Choudhury 2013). Thus, the present study focuses on consumer-based brand equity. Numerous studies have attempted to identify the fundamental drivers of brand equity (e.g., Chen 2010; Keller and Lehmann 2003; Park et al. 2010).

From an array of brand constructs, we choose to focus here on three fundamental brand equity drivers; namely, brand attachment, brand attitude, and brand awareness. Park et al. (2010) defined brand attachment as the feeling of self-connection, bond, or attachment towards a brand. Attachment to a brand is an emotional state involving the relationship between consumers and the brand (Louis and Lombart 2010). Brand attachment is important because it affects consumers' behavior and boosts brand profitability and consumers' lifetime value (Park et al. 2010). In other words, strong brand attachment results in resistance among consumers to changing to other brands and an enhanced ability of the brand to stand up to unexpected news or information (e.g., a product failure; Keller and Lehmann 2003). In the context of brand equity, it is important to predict psychological influences and their continuity, even in the face of an unclear future or fluctuating environments (Meyer 2012). Consistent with this, some scholars (e.g., Fedorikhin et al. 2008; Thomson et al. 2005) have emphasized the importance of emotion, which is a psychological attribute in brand attachment. Indeed, brand attachment concentrates on the connection between brand and consumer, so understanding it provides insight into consumer behavior, and especially purchase behavior (Meyer 2012). In addition, once consumers have become attached to a certain brand, they are more likely to purchase it (Meyer 2012) and pay a premium price for it (Thomson et al. 2005). Therefore, attachment is crucial in explaining the strength of the tie between consumer and brand, because attachment influences behaviors that enhance brand profitability (Thomson et al. 2005), resulting in higher brand equity. Accordingly, researchers have shown a significant interest in consumers' attachment to brands as a determinant of brand equity (Park et al. 2010).

Brand attitude refers to a consumer's overall evaluation of a brand in terms of both product and non-product attributes. These attributes include a set of established assumptions or feelings toward the brand, including brand appeal, likability, and favoritism (Spears and Singh 2004). Brand attitude is important because it forms the basis of consumers' behavior (e.g., choosing the brand), such that a positive consumer attitude toward a brand and a negative attitude toward competing brands affect their intention to purchase the brand (Laroche et al. 1996; Voester et al. 2016). As a result, some researchers have found that brands acquire more market share when consumer attitude toward them becomes more positive (e.g., Faircloth et al. 2001). Brand attitude is considered to be a vital factor because it is a foundation of consumer behavior (Keller 1993). Brand attitude can predict behaviors that are of interest to companies, such as purchase intention, purchase behavior, and brand choice (Priester et al. 2004). In addition, a consumer's brand attitude is considered to be representative of their overall associations with the

brand, and as a factor that brings about higher durability in consumer memory in relation to other associations (Chattopadhyay and Alba 1988). According to the above discussion, previous research in commercial branding has highlighted the significance of brand attitude and its possible antecedent relationship to outcome constructs such as brand equity (Faircloth 2001; Keller 1993; Lane and Jacobson 1995). Hence, Park et al. (2010) suggested that brand attachment and brand attitude are two essential drivers of brand equity.

Aaker (1991, p. 61) defined brand awareness as “the ability for a buyer to recognize or recall that a brand is a member of a certain product category”. There can be no brand if there is an absence of brand awareness (Wilcox et al. 2008), and brands that are better known among consumers are more likely to be included in consumers’ consideration set (Huang and Sarigöllu 2012). Consumers only choose brands of which they are aware (Shabbir and Rehman 2013); in fact, brand awareness plays a critical role in their final choice (Baker et al. 1986). In accordance with ANMT, Yasin (2009) claimed that brand awareness refers to a brand node in consumers’ memory that indicates the ability of consumers to recognize the brand among others in its category. In other words, brand awareness indicates how well a brand identity (e.g., a brand name) comes to consumers’ minds. Brand awareness is considered to be a significant factor in the consumer decision-making process, and plays a critical role in ensuring that consumers think about a certain brand when they envision a particular product category (Keller 1993). Put another way, increasing brand awareness enhances the likelihood that a specific brand will enter a consumer’s consideration set (i.e., range of brands contemplated for purchase) (Nedungadi 1990). Moreover, brand awareness could influence brand choice in the consideration set, even in the absence of other brand associations (Keller 1993). Many consumers tend to make purchase decisions only from familiar and well-established brands (Jacoby et al. 1977). In addition, in low-involvement buying decision processes, a minimum level of brand awareness will naturally help consumers to choose a specific product (Bettman and Park 1980). Accordingly, Keller and Lehmann (2003) claimed that brand awareness is one of the most important factors of the brand value chain, contributing to enhanced brand equity. This implies that brand awareness could be considered an important driver of brand equity. Overall, these constructs are highly relevant for strategic brand decisions (Eastman and Eastman 2011; Keller and Lehmann 2003; Park et al. 2010; Yasin 2009), which, in turn, influence brand equity (Yoo et al. 2000).

To sum up, brand attachment, brand attitude, and brand awareness are three interrelated constructs that reflect the domain of essential brand equity drivers. In Keller and Lehmann’s (2003) brand value chain model, brand attachment, brand attitude, and brand awareness, which are the sources of brand equity, are transferred to the next stage of the value chain; namely, brand performance. Taken together, these brand equity drivers influence how consumers will respond in the marketplace (Ambler et al. 2002). Ambler et al. (2002) noted six specific consequences of such reactions: price premium (i.e., how much extra consumers are willing to pay for an alternative product due to its brand), price elasticity (i.e., the extent to which consumers becoming more or less demanding when the price goes up or down), market share (i.e., the ability of marketing programs to build loyalty and drive brand

sales), brand expansion (i.e., the success of the brand in supporting category extensions and new product development in relevant categories), cost structure (i.e., the ability to cut marketing expenses due to prevailing consumer mindset), and brand profitability. Accordingly, in order to better understand how to manage and monitor consumer-based brand equity, we examine brand attachment, brand attitude, and brand awareness.

2.3 Brand image and brand equity

Since the 1950s, brand image has been considered a vital concept in marketing (Gardner and Levy 1955). Consistent with ANMT, Keller (1993, p. 3) defined the brand image as “perceptions about a brand as reflected by the brand associations held in consumer memory”. Brand associations are informal nodes that are linked in memory to the meaning of the brand for consumers, and play a vital role in achieving strong brand equity (Keller 1993). In other words, brand image describes the way consumers think about a brand and the feelings the brand brings to their minds when they think about it (Keller 2001). Therefore, a good brand image enables firms to achieve competitive advantage in the market (Roy and Banerjee 2007). From the end-users’ perspective, a strong brand image has a positive effect on consumers’ perception of the brand, which results in their being more likely to choose the brand over those competitors, and thereby makes the brand more competitive overall (Keller 1993). Moreover, Keller (1993) stated that a strong brand image affects the firm’s pricing strategy, promotion efforts, and distribution of the brand. Similarly, Kim et al. (2008) claimed that brand image has a significant effect on the brand’s marketing activities. According to Park et al. (1986), once a new product has acquired a distinct image relative to competing brands, its position in the market will become secure and its market share enhanced. However, some scholars have suggested that the significance of brand image in relation to consumers’ product assessment varies among product types, such that brand image may not be the main appraisal criteria for all products. For example, Graeff (1997) suggested that beer is frequently advertised and marketed mainly based on image, whereas automobile producers market their products based on non-image aspects, such as safety, warranty, price, and headroom.

Marketing researchers have proposed that brand image is a significant component of brand equity (e.g., Keller 1993). Consistent with this, a number of studies have conceptualized the direct impact of the brand image on brand equity (Faircloth et al. 2001; Krishnan 1996; Lassar et al. 1995). A summary of empirical studies on the relationship between brand image and brand equity is provided in Table 1. Several observations can be drawn from the review. First, most studies have examined the direct effect of brand image on brand equity. Thus, there is a lack of studies identifying the intervening mechanism that clarifies the path of influence from brand image to brand equity. Brand equity drivers potentially play a mediating role in connecting the brand image with brand equity, such that enhancing these drivers is believed to potentially increase brand equity (Keller and Lehmann 2003; Park et al. 2010). Brand equity drivers could serve as one of the mechanisms through which brand image influences brand equity, and their mediating role thus needs to be

Table 1 Empirical studies on the relationship between brand image and brand equity

Author(s)/year	Product/service category	Purpose of the study	Variables	Major findings
Yoo et al. (2000)	Athletic shoes, Camera film, and television sets	Exploring the relationship between retail brand image and brand equity	IV: retail brand image (or store image)	Retail brand image indirectly influenced brand equity through dimensions of brand equity (perceived quality and brand awareness)
Faircloth et al. (2001)	Sweaters	Examining the effect of brand attitude and brand image on brand equity	Mediator: dimensions of brand equity DV: brand equity IV: brand image, brand attitude	The results indicate that brand image is significantly influenced brand equity; however, brand attitude does not significantly affect brand equity
Chen (2010)	Electronics products	Examining the effect of green brand image on green brand equity	DV: brand equity IV: green brand image	(1) Green brand image significantly and positively influenced green brand equity (2) Green brand image significantly and positively influenced both green satisfaction and green trust (3) Both green satisfaction and green trust positively and significantly influenced green brand equity
Gill and Dawra (2010)	Toothpaste	Evaluating Aaker's sources of brand equity and the mediating role of brand image	Mediator: green satisfaction, green trust DV: green brand equity IV: brand awareness	Results indicated that brand awareness positively and significantly influenced the brand image. In addition, brand image positively and significantly affected brand equity

Table 1 continued

Author(s)/year	Product/service category	Purpose of the study	Variables	Major findings
Akbar and Azhar (2011)	Mobile phone	Investigating the impact of brand image on brand equity	IV: brand image	The authors showed that brand image has a significant positive effect on brand equity
Chahal and Bala (2012)	Healthcare sector	Examining the effect of perceived service quality, brand loyalty, and brand image on brand equity	DV: brand equity IV: perceived service quality, brand loyalty, brand image	The direct effect of brand image on brand equity was negative and insignificant. Yet, Brand image indirectly influenced brand equity through brand loyalty
Emari et al. (2012)	Chocolate	Examining the mediating impact of brand image on brand equity	Mediator: brand loyalty DV: brand equity IV: brand association, brand personality	Brand image is significantly and positively influenced brand equity. Moreover, the mediating role of the brand image between the brand association and brand personality as independent variables and brand equity as dependent variables was confirmed
Fatema et al. (2013)	Bank	Examining the effect of brand image and brand loyalty on brand equity	Mediator: brand image DV: brand equity IV: brand image, brand loyalty	Both brand loyalty and brand image significantly and positively influenced brand equity

Table 1 continued

Author(s)/year	Product/service category	Purpose of the study	Variables	Major findings
Ng et al. (2014)	Electric and electronic products	Analyzing the influence of brand perceive quality and credibility on consumer perceptions towards a brand green image, green value, and green equity	IV: perceived quality of brand Mediator: green brand perceived value, brand credibility, green brand image DV: green brand equity	Perceived quality of brand significantly and positively influenced the green brand image. Likewise, the green brand image is positively and significantly associated with green brand equity
Subramaniam et al. (2014)	Bank	Examining the effects of brand loyalty, image, and quality on brand equity	IV: brand loyalty, brand image, brand quality DV: brand equity	Brand image and brand loyalty has a significant positive effect on brand equity
Sasmitha and Mohd Suki (2015)	Clothing	Examining the effect of brand association, brand loyalty, brand awareness, and brand image on brand equity	IV: brand association, brand loyalty, brand awareness, brand image DV: brand equity	All brand association, brand loyalty, brand awareness, and brand image positively and significantly influenced brand equity
Liao et al. (2017)	Cosmetic products	Exploring the effect of brand personality on the relationship between antecedents of brand personality (e.g. brand awareness, brand association, perceived brand quality, brand image, and brand reputation) and brand equity	IV: antecedents of brand personality Mediator: brand personality DV: brand equity	The results showed that brand image significantly and positively influenced both brand personality and brand equity

In the review of empirical studies, only the variables related to this study are presented

IV Independent variable; DV dependent variable

explained, since exploration of potential mediators would provide better insights into how to leverage and attain organizational goals.

Second, prior studies have investigated the relationship between brand image and brand equity in the context of a single product/service type. For example, Faircloth et al. (2001) empirically tested the impact of brand image on brand equity in the sweater category. Likewise, Chen (2010) examined the relationship between green brand image and green brand equity in electronics products. Emari et al. (2012) assessed the impact of brand image on brand equity in chocolate. However, although the literature has suggested that brand image significantly influences brand equity, evidence demonstrating the moderating role of product type is lacking. The literature has sought to understand whether the strength of the relationship between brand image and brand equity differs among different product types. Consistent with this, previous studies (e.g., Chiang and Dholakia 2003; Fang 2012; Huang et al. 2009) have suggested that the impact of variables that influence consumer decisions to purchase a brand may differ depending on whether the products are search goods or experience goods. “Search goods are defined as those dominated by product attributes for which full information can be acquired prior to purchase; experience goods are dominated by attributes that cannot be known until purchase and use of the product or for which information search is more costly and/or difficult than direct product experience” (Klein 1998 p. 196). Consumers know that looking for information is costly and time-consuming, and that there exist trade-offs between the perceived costs and the advantages of additional searching (Stigler 1961). The cost of information searches may influence consumer decisions to purchase a brand, thereby affecting consumer-based brand equity. In this vein, Lovett et al. (2013) called for an extension to the literature in terms of exploring whether product type acts as a vital moderating construct. Accordingly, we use product type as a moderator.

Third, the literature has emphasized the value of WOM in the consumer decision-making process (Brown et al. 2007). WOM communication is generally considered to influence and inform consumers’ attitudes and behavioral intentions (Chevalier and Mayzlin 2006; Xia and Bechwati 2010). Therefore, the role of WOM in consumer behavior research comes to play. Consistent with this, many scholars have conducted research on the direct effect of WOM on consumer-related variables (e.g., Anwar Mir 2013; Chevalier and Mayzlin 2006; Herr et al. 1991; Jalilvand and Samiei 2012). Yet, in spite of the importance of WOM’s moderating role, it has largely been ignored. In this regard, while previous studies (e.g., Bambauer-Sachse and Mangold 2011) have investigated the direct effect of WOM on consumer-based brand equity, the moderating role of WOM has not, to our knowledge, been examined to date. We, thus take WOM as an additional moderating variable in our study.

We choose to focus on WOM for several reasons. First, WOM is considered an important force in the marketplace, influencing attitudes, preferences, purchase intention, and consumer decision making (Jalilvand and Samiei 2012; Wangenheim 2003). Accordingly, there has been a great deal of focus on WOM in the literature on relationship marketing as a potential reaction that can result from endeavors directed at shaping relationships with customers (Ismail and Spinelli 2012; Verhoef

et al. 2002). Indeed, firm profitability may be derived from the degree to which consumers engage in WOM, because they talk about their experience (positive or negative) with products or services with family, friends, colleagues, and others, which can affect the purchase behaviour of potential customers (Berry et al. 1994; Molinari et al. 2008; Zeithaml 2000), and in turn influences consumer-based brand equity. Second, when an individual thinks about a product, specific information that is most strongly linked to that product comes to their mind. This information comes from different sources, among which one of the most important is WOM. In other words, the literature has suggested that WOM is one of the sources that inform consumers' recall of a specific brand, whereby enhances brand awareness (Pitta and Katsanis 1995). Therefore, managers have become interested in WOM in the light of the fact that it is frequently an essential driver of consumer behavior, such as the adoption of a new product or the decision on which product to purchase. As a result, WOM may even be the main indicator of the product's success (Godes and Mayzlin 2004), leading to increased consumer-based brand equity.

Based on the above, our paper contributes to the existing literature in three ways. First, it examines the mediating role of brand equity drivers through which brand image may connect with consumer-based brand equity. Second, while previous studies have focused on a single product type, we examine the relationship between brand image and consumer-based brand equity along two prominent product types (i.e., search goods vs. experience goods). This should provide meaningful insights for marketers and managers through which they can understand whether the strength of the relationship between brand image and brand equity varies between different product types. Third, the present study extends previous works by building a conceptual framework that incorporates the moderating role of WOM communication effects on the relationship between brand image and consumer-based brand equity.

2.4 The effect of brand image on brand equity drivers

Brand attachment relates to consumers' psychological perception of a brand (Belaid and Behi 2011). In other words, brand attachment refers to consumers' experience of a brand. It prevents consumers from switching to other brands, increases consumers' forgiveness when facing negative information, and evokes consumer loyalty to the brand (Thomson 2006). Brand image has a similar effect, as it promotes thinking about a brand, and can increase excitement when thinking about it (Roy and Banerjee 2007). Therefore, both brand image and brand attachment are formed based on informational nodes linked to the brand nodes in the associative network memory. A positive brand image could create positive feelings towards the brand (Porter and Claycomb 1997), which, in turn, can result in greater emotional brand attachment (Malär et al. 2011). In addition, Belaid and Behi (2011) suggested that promoting brand image may lead to effective brand attachment. Accordingly, an established brand with a strong image will most likely lead to an enhanced emotional attachment to the brand. Thus, we propose that brand image has a positive effect on brand attachment.

Brand attitude underlies consumer behavior and brand choice. Consumers with a positive attitude toward a brand are willing to pay premium prices for the brand (Keller 1993). Therefore, a significant aspect of the brand image that influences consumers' responses to prices is the overall brand attitude. Keller (1993) stated that brand attitude is about (1) consumers thinking about a brand's attributes and benefits, and (2) judgment of their beliefs. It is therefore reasonable to predict that brand image could have a direct effect on brand attitude. Porter and Claycomb (1997) claimed that brand image could produce a positive attitude towards the brand. Moreover, Kim (2000) argued that promoting brand image could result in creating a strong brand attitude. Hsieh et al. (2004) also suggested that brand image should include overall brand attitude. Specifically, a positive brand image could encompass positive brand attitudes. Therefore, we predict that brand image has a positive effect on brand attitude.

Brand awareness influences which brands are considered by consumers, and also affects which brands are chosen from the consideration set. Thus, brand awareness has a significant effect on the consumer decision-making process (Macdonald and Sharp 2000). In fact, brand awareness is related to the probability that a brand name comes to consumers' minds easily (Keller 1993). Prior research has examined the impact of brand awareness on brand choice (e.g., Macdonald and Sharp 2000), but not many researchers have studied the predictors of brand awareness. More specifically, extant studies (e.g., Keller 1993; Esch et al. 2006) have only considered the effect of brand awareness on brand image, and not the reverse. This suggests that the popularity of a certain brand (i.e., brand awareness) leads to attractiveness of the brand (i.e. brand image). On the contrary, Padilla and Watson (2010) claimed that brand image may also influence brand awareness. Consistent with this, Djakeli (2012) stated that a robust brand image is an important component for building brand awareness. Indeed, creating a positive brand image may lead consumers to develop brand perceptions toward a specific brand through others' reactions with regard to the brand (Charumbira and Stotlar 2015). That is, positive actions by others towards a specific brand (e.g., positive WOM) may influence consumers to develop favorable associations for the brand. This may convey brand information to other consumers, and thereby increase brand awareness (Charumbira and Stotlar 2015). In addition, Kayaman and Arasli (2007) claimed that brand image may increase the probability of brand choice, and hence consumer brand awareness.

Therefore, this study asserts that a strong brand image will increase the likelihood of brand identification. This, in turn, is related to consumer buying behavior, particularly the identification of alternative choices among favorable brands that exhibit a strong image. As a result, the brand image may have a positive effect on brand awareness. To sum up, we propose that brand image has a positive effect on brand attachment, brand attitude, and brand awareness. Thus, we propose the following hypotheses:

- H1 Brand image has a significant positive effect on brand attachment.
- H2 Brand image has a significant positive effect on brand attitude.
- H3 Brand image has a significant positive effect on brand awareness.

2.5 The effects of brand image and respective drivers on brand equity

The preceding hypotheses suggest a relationship between brand image and brand equity drivers. This section aims to show how these focal brand constructs relate to consumer-based brand equity. Brand image, which is founded in associative network memory, is crucial to consumer decision making and consumers' evaluation of a specific brand (Faircloth et al. 2001), and may contribute to building strong brand equity (Holden 1992). Brand associations, which are obtained via marketing mix activities, are important factors in building brand image. Scholars have claimed that building brand equity is partially related to the nature of brand associations that create the brand image (Faircloth et al. 2001). In addition, Pitta and Katsanis (1995) suggested that a unique, favorable, and strong brand image will remain in the consumer's mind, thereby contributing to increased brand equity.

Scholars have proposed that brand image is a critical component of brand equity (Keller 1993). Krishnan (1996) indicated that brands with high equity are more likely to have a more positive brand association (brand image) than those with low equity. In addition, Lassar et al. (1995) suggested that higher brand image is associated with higher brand equity. Moreover, Faircloth et al. (2001) found that brand image has a significant positive influence on brand equity. Thus, we hypothesize:

H4 Brand image has a significant positive effect on brand equity.

Individuals with a strong attachment to the brand desire to help the organization to succeed (Fullerton 2005). Accordingly, Park et al. (2010) claimed that attachment to a brand is important because it influences consumers' behavior, leading to increased profitability and consumer lifetime value. Moreover, Thomson (2006) suggested that consumers' feelings that are linked to brand attachments are the basis of strong brand relationships. Thomson et al. (2005) claimed that strong brand attachment may relate to investment in the brand. For example, consumers may be willing to pay a premium price to attain the brand. Thus, we can assume that brand attachment has a positive effect on brand equity.

Faircloth et al. (2001) and Buil et al. (2009) specified that brand equity is affected by consumers' attitude toward a brand. In addition, Faircloth et al. (2001) stated that the market share of two-thirds of brands will increase when consumers' attitudes toward these brands becomes positive. Keller (1993) also suggested that consumers who have a strong attitude toward a brand are more willing to pay a premium price for it. If consumers have a positive attitude toward a brand, there is a greater likelihood that the brand will be preferred or considered for purchase. Thus, it can be assumed that positive attitude leads to high consumer-based brand equity.

Huang and Tsai (2013) suggested that brand awareness is one of the factors that companies with a brand orientation enjoy. Brand awareness is one of the most important components in creating brand equity (Seno and Lukas 2007); hence, it is reasonable to propose that a high level of consumer awareness toward a brand may lead to high consumer-based brand equity. More specifically, brands exhibiting a higher degree of awareness warrant consumer confidence and familiarity; thus, they should be associated with higher brand equity. Taken together, we assume that all

three drivers of brand equity have a positive impact on brand equity. Thus, the following hypotheses are put forth:

- H5 Brand attachment has a significant positive effect on brand equity.
- H6 Brand attitude has a significant positive effect on brand equity.
- H7 Brand awareness has a significant positive effect on brand equity.

2.6 Moderating effect of product type

Product classification is a significant research aspect in consumer behavior studies because consumer behavior varies with attributes of certain product types (Bei and Widdows 2004). Accordingly, Nelson (1974) divided products into two groups based on a simplified theory of consumers' quest for quality products: search goods and experience goods. Search goods are those for which consumers can determine the level of quality prior to purchase via inspection (e.g., shoes, apparel), whereas for experience goods, consumers cannot determine the quality prior to purchase (e.g., shampoo, deodorant). To the best of our knowledge, previous researchers have only examined consumer-based brand equity in relation to a specific product category or service. For example, Kayaman and Arasli (2007) and Kim et al. (2008) examined brand equity in the context of hospital services. Faircloth et al. (2001) found that a positive brand image can significantly increase brand equity in sweaters. They also indicated that brand attitude does not significantly influence brand equity. However, among different product types, the effect of the brand image may differ. Lovett et al. (2013) examined the characteristics of brands, and mentioned that product type may play an important moderating role. In this study, we focus on clothing, shoes, handbags, and wallets as search goods; and deodorant, shampoo, clocks, and watches as experience goods. We selected these products because they are frequently used by Malaysian consumers.

Individuals who have greater familiarity with the brand, generated through direct or indirect brand-related experiences, tend to be more knowledgeable about the brand and its attributes (Alba and Hutchinson 1987). King and Balasubramanian (1994) found that when assessing search goods, consumers are more likely to employ their own decision-making strategies compared to when they are assessing experience goods. This is because it is difficult to assess the quality of experience products prior to purchase, and individuals are more likely to accept product recommendations for these products than for search goods (Senecal and Nantel 2004; Sellers-Rubio and Más-Ruiz 2015). In addition, buyers of experience products are less likely than those of search products to engage in free-riding behavior (Huang et al. 2009). The reason for this is that since individuals put greater effort into assessing experience attributes, the initial cost of gaining experience information is higher (Huang et al. 2009).

Moreover, one of the main advantages of pre-purchase searching is that it reduces the perceived risk associated with buying a product. For search goods, consumers are highly confident in the acquired information prior to purchase; on the contrary, information about experience goods is subjective, and rooted in consumer

judgments and beliefs of individuals (Wright and Lynch 1995). Recognizing that different types of products encompass different consumer behavioral aspects (e.g., product assessment, information search, perceived risk, consumer involvement) when arriving at a purchase decision, we predict that there is a difference between types of products in estimating consumer-based brand equity. Thus, we propose that:

H8 Product type has significant moderating effects on the relationships between brand image, drivers of brand equity, and brand equity.

2.7 Moderating effect of WOM

The significance of WOM in marketing is not a new phenomenon (Keller 2007). In other words, WOM may be the oldest way of exchanging opinions on different goods and services offered by markets (Goyette et al. 2010). According to Arndt (1967), WOM refers to oral communication among people regarding a brand, product, or service. WOM is one of the most influential sources of information transferred by consumers (Duana et al. 2008), and encompasses well-documented advantages (Hoch and Ha 1986). Research on WOM has indicated that consumers rely on recommendations from other people in purchase decision-making; that people are motivated to spread their opinion about a product; and that people have different degrees of impact on their friends, colleagues, and families in WOM communication (Trusov et al. 2009). Researchers have found that WOM has a significant effect on consumers' decision-making process, in terms of increasing acceptance of a new product when WOM on that product is positive (Sheth 1971), and switching away from a brand when they receive negative information about it (Bone 1992). Indeed, individuals may reduce the risks associated with their purchases by looking for information from reliable sources, such as friends or those who have relevant expertise (Cobb-Walgren et al. 1995).

A number of studies have demonstrated that WOM communication often significantly influences judgments on brands. For example, Engel et al. (1969) found that individuals often rely on WOM when choosing an automotive diagnostic center. Feldman and Spencer (1965) also indicated that people utilize WOM when choosing a physician. Furthermore, consumers trust WOM when assessing the potential purchase of a new product or service (Brown and Reingen 1987). Other researchers have specified that WOM affects brand attitude (e.g., Herr et al. 1991; Anwar Mir 2013) and brand awareness (Pitta and Katsanis 1995). Brand success relies on creating bonds (e.g., attachment, familiarity, etc.) with customers. These customers engage in multiple social networks, where they influence one another through WOM, shaping attitudes and behaviors that can strengthen or weaken these bonds (Allsop et al. 2007). Therefore, WOM has become an important benchmark for capturing consumers.

Srivastava (2009) claimed that the buildup of brand equity depends on consumers holding the brand in their mind. WOM plays a critical role in shaping consumer perceptions (Plummer 2007). However, WOM could have both a positive and negative influence on consumers' perception towards a brand (Lim and Chung 2011). Positive WOM enhances the likelihood of purchasing, whereas negative

WOM results in diffusion of complaints, and yields a detrimental effect (Xu and Chan 2010). Thus, in the present study, we predict that WOM plays a moderating role on the relationships among brand image, drivers of brand equity, and consumer-based brand equity. It can be inferred that the effects on the drivers of brand equity, as well as brand equity itself, depend on whether consumers engage heavily in WOM. Formally, we predict that:

H9 WOM has significant moderating effects on the relationships between brand image, drivers of brand equity, and brand equity.

The conceptual framework comprising H1 to H9 is illustrated in Fig. 1. The present study postulates that the brand concepts (i.e., brand image, brand attachment, brand attitude, brand awareness, and consumer-based brand equity) are interrelated and represent different aspects of an individual's brand perception and processing. As shown in Fig. 1, brand image positively influences brand equity directly, as well as through brand equity drivers, suggesting that brand equity drivers partially mediate the relationship between brand image and brand equity. A superior brand image can be distinguished from others and is firmly positioned in the consumers' mind, and thereby has potential to increase the brand equity (Pitta and Katsanis 1995).

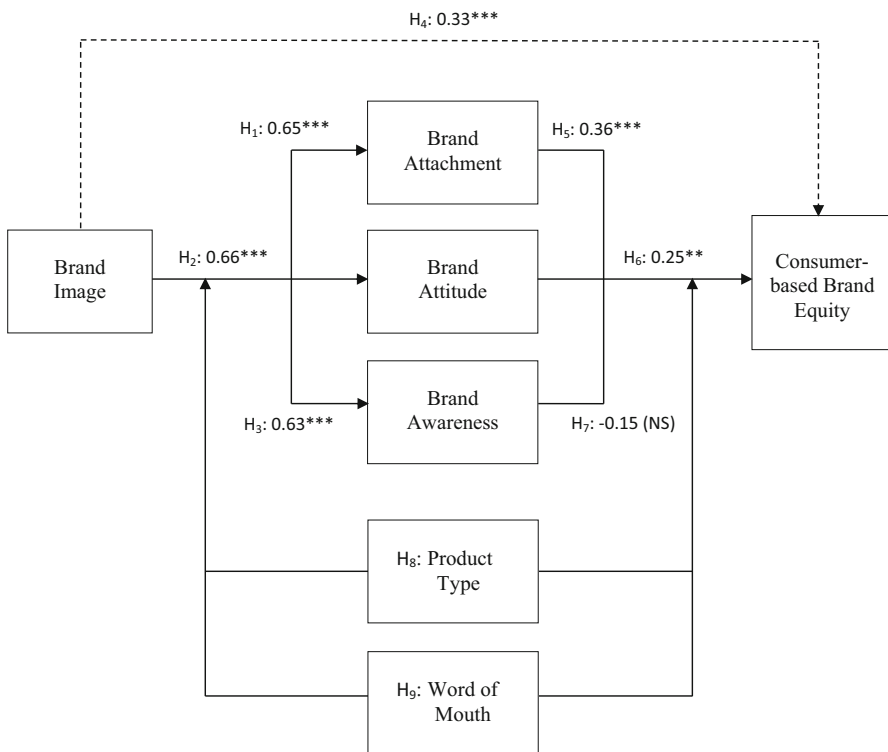


Fig. 1 The relationships between brand image and brand equity. * $p < .05$; ** $p < .01$; *** $p < .001$; NS not significant

Brand image is associated with consumer experience with the brand (Hung 2008). Correspondingly, brand attachment is formed based on consumers' experiences with the brand, and prevents consumers from switching to other brands even when unfavorable information is received (Thomson 2006). Therefore, a positive brand image can increase brand equity through brand attachment.

Brand image was established from the concept of associative network memory, and is vital for consumer brand evaluations and building brand equity (Faircloth et al. 2001). Consumers capitalize on product cues to create beliefs about a product, which in turn shapes their purchase behavior. In general, brand image is considered to be made up of product cues (Cordell 1992; Thorelli et al. 1989), and consumers shape stereotypical beliefs about products from particular brands and the features of products (Yasin et al. 2007). Consequently, brand image has the ability to evoke consumers' beliefs about product features, and influences their evaluations thereof (Srikatanyoo and Gnoth 2002). This implies that positive brand image fosters positive brand attitude among consumers, and this eventually results in enhanced brand equity.

A strong brand image could increase brand equity as it enables the brand to command larger margins (Keller 1993). However, a brand that is retrieved early and recalled easily will have a stronger impression in the consumer's consideration set (Alba and Chattopadhyay 1986), which in turn influences product purchase (Vieceli and Shaw 2010). A brand with a positive image that is recalled prior to other brands of the same category may have more positive brand equity compared to competing brands.

2.8 Control variables

To investigate the effect of extrinsic factors on the relationships between the research constructs, we incorporate several control variables into our research model and analysis. These include gender, age, income, and price consciousness. Price consciousness is defined as the extent to which buyers concentrate on searching and paying a low price for a product or service (Kukar-Kinney et al. 2012).

3 Methodology

3.1 Sample and data collection

We collected data using a structured survey questionnaire. The questionnaires were administered to students in Malaysia based on the convenience sampling method. Respondents were initially requested to choose a consumer product and identify the name of a brand based on a product category of their choice (i.e., search goods, namely clothing, shoes, handbags, and wallets; and experience goods, namely deodorant, shampoo, clocks, and watches). The statements for all multiple-item constructs were based on a five-point Likert scale (anchored by 1 = "strongly disagree" to 5 = "strongly agree"). Participants were asked to answer the questionnaire based on the brand they identified earlier. The participants provided demographic information (i.e., gender, age, monthly income, and education level) in

the last section of the questionnaire. A pre-test with a small sample ($n = 30$) of experts in the area and population under study was conducted. The participants were asked to provide their opinion about the overall structure of the questionnaire and problems relating to the clarity of the items and instructions. We adapted the items in the questionnaire based on the feedback received from the pre-test.

According to Bentler and Chou (1987), a sample size can be as low as a ratio of 5:1 between the number of responses and items in the questionnaire. Accordingly, we administered 352 questionnaires, assuming that some would be returned incomplete; of these, 321 were usable. Among the participants, 117 were male (36%) and 204 were female (64%); 115 (36%) were aged 20 or below, 166 (52%) were between 21 and 30, 32 (10%) were between 31 and 40, and 8 (2%) were 41 years old or older. Concerning monthly income, 231 respondents (72%) had a monthly income of 1000 RM (Ringgit Malaysia) or below (at the time of data collection, 1 USD = 3.2 RM), 45 (14%) had a monthly income of 1001–2500 RM, 30 (9%) had a monthly income of 2501–4000 RM, and 15 (5%) had a monthly income of 4001 RM or higher. In terms of education level, 25 participants (8%) had a diploma or below, 193 (60%) had a bachelor's degree, and 103 (32%) had a master's degree or above.

3.2 Measures

All items in the questionnaire, except for those on demographic variables and product type, were measured based on a five-point Likert scale. The participants were requested to indicate their opinion (anchored by 1 = “strongly disagree” to 5 = “strongly agree”) to a series of statements (shown in Table 1) about the brands they had identified. For brand image, we adapted the measurements from Martin and Brown (1990), Aaker (1996), and Weiss et al. (1999). The items included functional image, affective image, and reputation image. Brand attachment included four items used by Louis and Lombart (2010). The brand attitude measures were adopted from Spears and Singh (2004). For the brand awareness construct, we adopted measures from Oh (2000). For brand equity, we employed three items that were used in He and Li's (2011) research. The measurement for WOM was adapted from Lam et al. (2009) and Vázquez-Casielles et al. (2013). To measure price consciousness, which served as a control variable, we adopted items from Kukar-Kinney et al. (2012). Following Nelson (1974), products were classified into two groups, namely search goods and experience goods. We coded “search goods” as 0 and “experience goods” as 1 in our analysis.

4 Results

4.1 Validity and reliability

In order to refine the measures and assess the validity of the constructs, Confirmatory Factor Analysis (CFA) was used for the entire set of constructs. Two items were removed due to low factor loadings (less than .50; see Table 2),

Table 2 Confirmatory factor analysis: construct measurement summary

Construct and item description	Standardized factor loadings
<i>Brand image</i> ($\alpha = .84$, CR = .86, AVE = .51, HSIC = .48)	
1. This brand has a high quality	.74
2. This brand has better characteristics than its competitors	.74
3. This brand has a personality that distinguishes itself from competitors	.69
4. This brand is a brand that does not disappoint its customers	.65
5. This brand is one of the best brands in the sector	.73
6. This brand is stable in the market.	.70
<i>Brand attachment</i> ($\alpha = .83$, CR = .84, AVE = .57, HSIC = .55)	
1. I love this brand very much and care about it	.80
2. I am connected to this brand	.67
3. I am attracted to this brand	.84
4. Thinking about this brand brings me a lot of pleasure	.69
<i>Brand attitude</i> ($\alpha = .89$, CR = .90, AVE = .64, HSIC = .60)	
1. I feel that this brand is appealing	.73
2. I feel that this brand is good	.86
3. I feel that this brand is pleasant	.81
4. I feel that this brand is favorable	.81
5. I feel that this brand is likable	.77
<i>Brand awareness</i> ($\alpha = .88$, CR = .89, AVE = .67, HSIC = .48)	
1. This brand name is familiar	.78
2. This brand name is well-known.	.91
3. This brand name is visible	.67
4. This brand name is famous	.87
<i>Word of mouth</i> ($\alpha = .75$, CR = .81, AVE = .51, HSIC = .24)	
1. I share my information about this brand with people	.67
2. I seek out the advice of people before I choose this brand	.74
3. The probability of accepting advice from other people is high	.72
4. The probability of choosing this brand before receiving recommendations is low ^a	
5. The probability of choosing this brand after receiving recommendations is high	.71
<i>Brand equity</i> ($\alpha = .81$, CR = .81, AVE = .59, HSIC = .47)	
1. It makes sense to buy this brand instead of other brands, even if they are the same	.73
2. Even if another brand has similar features, I would prefer to buy this brand	.81
3. If there is another brand as good as this brand, I prefer to buy this brand	.76
<i>Price consciousness</i> ($\alpha = .72$, CR = .75, AVE = .51, HSIC = .12)	
1. I check the price even for inexpensive brands before buying	.50
2. I read price tags of the products I buy ^a	
3. Low price is an important consideration in my purchases	.83

Table 2 continued

Construct and item description	Standardized factor loadings
4. No matter what I buy, I shop around to get the lowest price	.76

All constructs were measured with a five-point Likert scale: 1 = Strongly Disagree, 5 = Strongly Agree
CR composite reliability, *AVE* average variance extracted, *HSIC* highest squared inter-construct correlation

^a Item was dropped during the scale purification process

which resulted in a satisfactory fit to the data. Table 2 shows a summary of CFA results.

The results show that our measurement model fitted the data adequately: $\chi^2 = 608.53$ ($p < .001$); d.f. = 352; $\chi^2/\text{d.f.} = 1.73$, normed fit index (NFI) = .90, goodness of fit index (GFI) = .90, incremental fit index (IFI) = .95, comparative fit index (CFI) = .95, and root mean square error of approximation (RMSEA) = .05. These results show that the proposed model could be accepted (Hair et al. 2010).

To assess the validity of the constructs, we used convergent and discriminant validity methods. As indicated in Table 1, the average variance extracted (AVE) of each construct was higher than .50. Accordingly, the measures show adequate convergent validity (Kamdar and Dyne 2007). Concerning discriminant validity, we compared squared inter-construct correlations for pairs of constructs with the AVE of each construct. As indicated in Table 2, all AVE scores were larger than the corresponding highest squared inter-construct correlation estimates. Therefore, there is sufficient evidence for discriminant validity (Kamdar and Dyne 2007).

To assess the internal consistency of the constructs, we used the Cronbach's alpha coefficient. As shown in Table 2, the Cronbach's alpha of all constructs was higher than .70. Additional support for the reliability of the constructs, specifically composite reliability (CR) scores, is also shown in Table 2. All CR scores were larger than .70. Accordingly, the constructs can be said to demonstrate adequate internal consistency (Fornell and Larcker 1981; Nunnally 1978).

4.2 The relationship between brand image and brand equity

To examine the hypothesized relationships, we performed Structural Equation Modeling (SEM) with the maximum likelihood estimation method using the model without moderators. Based on our CFA results, the model was tested with standardized coefficients and other fit statistics. To assess the differential effects, standardized coefficients as path coefficients are reported. First, the overall model fit was examined. The Chi square ($\chi = 412.327$, degrees of freedom [d.f.] = 200) is significant at the .001 level. The fit indices are as follows: NFI = .90, IFI = .95, CFI = .95, RMSEA = .05. According to Hair et al. (2010), it can be concluded that the model fitted the data reasonably well.

H1, H2, and H3 examine the effect of brand image on brand attachment, brand attitude, and brand awareness, respectively (Table 3). As indicated in Table 3, all

Table 3 Standard coefficient and t-value for each path

Number	Hypotheses	Standard coefficient	T-value
1	Brand image → Brand attachment	.65***	10.58
2	Brand image → Brand attitude	.66***	17.05
3	Brand image → Brand awareness	.63***	13.06
4	Brand image → Brand equity	.33***	4.05
5	Brand attachment → Brand equity	.36***	6.05
6	Brand attitude → Brand equity	.25**	2.54
7	Brand awareness → Brand equity	-.15 (NS)	-2.20
<i>Control variables</i>			
1	Gender → Brand equity	.05 (NS)	.64
2	Age → Brand equity	-.04 (NS)	-.64
3	Income → Brand equity	-.03 (NS)	-.53
4	Price Consciousness → Brand equity	.03 (NS)	.61

* $p < .05$; ** $p < .01$; *** $p < .001$; NS not significant

three paths were highly significant ($p < .001$), indicating a positive effect on brand attachment ($\gamma = .65$, $t = 10.58$), brand attitude ($\gamma = .66$, $t = 17.05$), and brand awareness ($\gamma = .63$, $t = 13.06$). Accordingly, H1, H2, and H3 receive strong support.

To test H4 to H7, both product image and brand equity drivers were set as independent constructs, and brand equity was set as the dependent construct. The results shown in Table 3 indicate that brand image had a significant positive effect on brand equity ($\gamma = .33$, $t = 4.05$) at the significance level of .001. Therefore, H4 is supported. Brand attachment ($\gamma = .36$, $t = 6.05$) and brand attitude ($\gamma = .25$, $t = 2.54$) had significant positive effects on brand equity ($p < .001$, $p < .01$, respectively). Therefore, H5 and H6 are supported. The path from brand awareness to brand equity, however, was not significant ($\gamma = -.15$, $t = -2.20$, $p > .1$). Thus, H7 is not supported. These results suggest that to attain stronger brand equity, firms should concentrate on improving brand attachment, brand attitude, and brand image, rather than intensifying brand awareness. According to Fig. 1, brand attachment and brand attitude partially mediate the relationship between brand image and brand equity. In specific terms, brand image positively affects brand equity through brand attachment and brand attitude. None of the control variables (i.e., gender, age, monthly income, and price consciousness) had a significant effect on brand equity (Table 3).

4.3 The moderating effects of product type and WOM

One widely used procedure to test moderating effects in SEM involves splitting the scale at a certain point and assigning the values above and below that point as two groups (MacCallum et al. 2002; Im et al. 2008). H8 and H9 examine the moderating effects on the relationships among brand image, drivers of brand equity, and brand equity. Because product type is a categorical construct, we used multi-group

analysis with AMOS to test H8. We conducted the test in a two-step procedure. First, the appropriate structural parameters were constrained to be equal across groups, which resulted in producing an estimated covariance matrix for each group and a total χ^2 value for the sets of sub-models as the section of a single structural system. The parameter equality constraints were then eliminated, which led to generating a second χ^2 value with lower degrees of freedom. The moderating effects were tested by evaluating whether there is a statistical difference between the two χ^2 values. If the change in χ^2 value, normally a reduction, is statistically significant, a moderator effect exists (Brockman and Morgan 2003; Im et al. 2008). These two steps solved the issue of the difference between the two models being affected by error variances and factor loadings (Zhang et al. 2013). Once the moderating effect has been examined and the significant differences between the two groups considered, the standard coefficients between the two groups should be compared (Dabholkar and Bagozzi 2002). If the change is higher than .1, this means that the difference is at a highly significant level (S^{++}). If the change is higher than .05 and less than .1, this means that the difference is at a significant level (S^+). If the change is less than .05, this indicates that there is no significant difference (NS). However, if the path is at a significant level for one group and not for another, the difference is at a significant level (S^+).

As indicated in Table 4, there were significant differences between search goods and experience goods ($\Delta\chi^2 = 34.43, p < .05$). This suggests that product type could have acted as a moderator; thus, H8 is supported. Brand image strongly influenced brand attachment for both product types, with no significant difference. Brand image had a significant effect on brand attitude for both product types, and its influence was stronger for search goods than for experience goods ($\gamma_{\text{search goods}} = .70 > \gamma_{\text{experience goods}} = .64$). The brand image also significantly influenced brand awareness for both product types. Although brand attachment had a significant effect on brand equity for both product types, the effect was much higher for experience goods than for search goods ($\gamma_{\text{experience goods}} = .46 > -\gamma_{\text{search goods}} = .25$). Brand attitude had a highly significant effect on brand equity for both product types. However, its effect on brand equity was much higher for search goods than for experience goods ($\gamma_{\text{search goods}} = .61 > \gamma_{\text{experience goods}} = .31$). Finally, brand awareness had no significant effect on brand equity for both product types.

Because WOM is not a categorical construct, we split the groups into high and low levels using the median split approach (Baron and Kenny 1986; MacCallum et al. 2002) to test H9. The results (Table 4) indicate that there was a significant difference between low and high levels of WOM ($\Delta\chi^2 = 40.94, p < .05$). Thus, WOM could have acted as a moderator, and thereby H9 receives support. Brand attachment had a significant effect on brand equity for both low and high levels of WOM, with no significant difference. Brand image significantly affected brand attachment for both groups of WOM; however, its effect on brand attachment for people who engaged in the levels of WOM was much higher than for those who engaged in high levels of WOM ($\gamma_{\text{low}} = .69 > \gamma_{\text{high}} = .50$). In addition, brand image was found to significantly influence brand attitude, with a significant difference between the two levels of WOM. Although the brand image had a

Table 4 Comparison of the coefficients between search goods and experience goods, and low and high levels of WOM

Number	Hypotheses	Product type		WOM			
		Standard coefficient		t-value		Standard coefficient	
		Search goods	Experience goods	Search goods	Experience goods	Low	High
1	Brand image → Brand attachment	.65***	.65*** (NS)	10.53	10.53	.69***	.50*** (S ⁺⁺)
2	Brand image → Brand attitude	.70***	.64*** (S ⁺)	12.77	13.31	.69***	.59*** (S ⁺⁺)
3	Brand image → Brand awareness	.68***	.62*** (S ⁺)	8.72	10.24	.65***	.57*** (S ⁺)
4	Brand attachment → Brand equity	.25*	.46*** (S ⁺⁺)	2.36	6.36	.44***	.40*** (NS)
5	Brand attitude → Brand equity	.61***	.31*** (S ⁺⁺)	3.83	2.86	.32***	.48*** (S ⁺⁺)
6	Brand awareness → Brand equity	-.15	-.02 (S ⁺⁺)	-1.34	-.28	-.02	-.10 (S ⁺)

Multigroup analysis for product type: $\Delta\chi^2 = 34.43$, $p < .05$

Multigroup analysis for WOM: $\Delta\chi^2 = 40.94$, $p < .05$

NS no significant difference; S⁺ significant; S⁺⁺ highly significant

* $p < .05$; ** $p < .01$; *** $p < .001$

significant effect on brand awareness, its influence was stronger in the low level of WOM compared to in the high level of WOM. Brand attitude significantly influenced brand equity for both low and high levels of WOM. The effect of brand attitude on brand equity was much higher for individuals who were highly engaged in WOM than for those who had little engagement in WOM ($\gamma_{\text{high}} = .48 > -\gamma_{\text{low}} = .32$). Brand awareness did not significantly affect brand equity for either the low or the high level of WOM engagement.

5 Discussion

5.1 Summary

The results captured in this study show the usefulness of our conceptual model in understanding the link between brand image and brand equity. The results indicate that brand image has a positive effect on brand attachment, brand attitude, and brand awareness. This is consistent with previous researchers' suggestions (Belaid and Behi 2011; Kayaman and Arasli 2007; Porter and Claycomb 1997). The results are also consistent with previous findings (Faircloth et al. 2001; Wilcox et al. 2008) that brand image appears to have a strong influence on brand equity.

The results of this study verify the mediating role of brand attachment and brand attitude. These constructs were found to partially mediate the effect of brand image on brand equity. According to Frazier et al. (2004), a mediator clarifies how or why a predictor leads to its outcome construct. Our findings imply that positive perceptions about a brand are critical, and are likely to help consumers positively evaluate a brand, and enhance their attachment towards it. Based on our findings, we can conclude that brand attachment and attitude are important elements relative to brand awareness in order to generate strong brand equity.

The results confirm that the brand concepts in the present study (i.e., brand image, brand attachment, brand attitude, brand awareness, and consumer-based brand equity) are interrelated. Brand image boosts brand equity, and this is mediated by brand attachment and brand attitude. Brand image, attachment, and attitude are prerequisites of consumer-based brand equity. Therefore, developing a positive image, strengthening brand attachment, and creating an affirmative attitude among consumers should be synchronized to enhance consumer-based brand equity. In addition, brand image is an antecedent of brand attachment, brand attitude, and brand awareness.

In the present research, the moderating roles of product type and WOM on the relationship between brand image and consumer-based brand equity were investigated. Although brand image was found to significantly influence brand attachment for both product types, there was no significant difference between search goods and experience goods. Moreover, we found that brand image affected brand attitude and brand awareness across both product types, with a significant difference between search goods and experience goods. Indeed, the effect was higher for search goods than for experience goods. That is, promoting brand image could result in generating a higher level of positive brand attitude and brand

awareness among search goods compared to experience goods. The results also indicate that brand attachment has a significant influence on brand equity for both product types, with a higher effect for experience goods than for search goods. Based on our findings, the effect of brand attitude on brand equity is much stronger for search goods than for experience goods. Our results show that brand awareness does not significantly influence brand equity; this applies to both search and experience goods. Some scholars (e.g., Keller 1993; Kayaman and Arasli 2007) have previously emphasized the importance of brand awareness in generating brand equity. Although awareness of a brand is fundamental for it to be part of consumers' consideration set, this research suggests that it is not a key component of brand equity. Our result is consistent with the study of Bailey and Ball (2006) and Kayaman and Arasli (2007), who also found that brand awareness is not an essential element to building brand equity in their respective areas. This implies that simply having a brand name does not ensure success in the marketplace (Bailey and Ball 2006); other factors, such as developing a positive brand attitude and brand attachment are important in fostering brand equity.

WOM appears to significantly moderate the effect of brand image on brand attachment, brand attitude, and brand awareness. Similarly, the impact of brand attitude on consumer-based brand equity is significantly moderated by WOM. The results show, however, that WOM does not moderate the relationship between brand attachment and brand equity. This is perhaps partly due to the fact that brand attachment is formed based on individuals' perceptions of a brand (Belaid and Behi 2011), rather than acceptance of recommendations from others. An interesting finding is that although brand awareness does not significantly influence brand equity, this relationship is significantly moderated by WOM.

5.2 Contributions to scholarship

Theoretically, this study contributes to the literature by offering a comprehensive model with multiple branding constructs that examine direct and contingent relationships. Our study contrasts with past empirical research in that we suggest that the relationship between brand image and brand equity is context specific. We believe that considering multiple brand constructs, coupled with product type and WOM as moderators, provides more insightful information to understand brand equity. In short, this study broadens the conceptualization of the impact of brand image on brand equity.

5.3 Managerial implications

This study provides a number of important practical implications for marketers and managerial decision-making. First, the study highlights the importance of brand image and equity for current businesses. In today's competitive business world, it is difficult for companies to differentiate themselves from their competitors. Technology leaders typically view business growth through product proliferation and innovation as a major challenge, as the present market is occupied by rivals with very little product differentiation. Firms need to search for ways to remain

competitive in the marketplace. One way to achieve this goal is to create a positive brand image in the consumer's mind. Although there are numerous factors that may affect consumers' purchase decision-making process, brand image is doubtless highly relevant to managers, as it significantly increases consumers' brand attachment, attitude, and awareness (as shown in this study). Not only are these elements important for brand recall, but they also allow such brands to be considered in future purchases (Wilcox et al. 2008). This, in turn, increases brand equity.

Second, this study implies that brand image and brand equity are moderated by product type and WOM. When managers examine the effect of brand image on brand equity, product type and WOM should be taken into consideration. The findings of this study suggest that firms selling search goods, such as Adidas and Nike, should concentrate more on promoting their brand image, compared to those selling experience goods, such as Head & Shoulders and Pantene, in order to reinforce brand attitude and awareness. Therefore, the present study suggests that the effect of brand image is more pertinent for products that have search properties (e.g., color, size, and style) rather than those with experience properties (e.g., the cleansing and lathering properties of shampoo). In other words, firms selling search goods are more likely to increase their brand attitude and brand awareness through building a positive brand image. In addition, in terms of brand equity drivers, brands selling search goods, such as Adidas and Nike, should devote greater attention to developing positive brand attitude among prospective buyers, while those selling experience goods, such as Head & Shoulders and Pantene, should focus more on reinforcing brand attachment in order to build strong consumer-based brand equity.

Apart from that, it is apparent that managers have low influence on WOM, but should be aware of its impact on branding effectiveness. Our study indicates that the effect of brand image on drivers of brand equity is much stronger when consumers engage in a low volume of WOM. Put another way, the effect of consumers' overall perception of the brand on drivers of brand equity is weaker when consumers engage in a high level of WOM. This may be partly due to the fact that consumers also share unsatisfactory experiences with the brand when participating in high levels of WOM (Collier 1995; Tong et al. 2013), which, in turn, has an adverse effect on consumer behaviors and brand attitude. Conversely, the impact of brand attitude on consumer-based brand equity is stronger when consumers engage in high levels of WOM. Therefore, by managing brand image and WOM communications, marketers can take advantage of drivers of brand equity (i.e., brand attachment, brand attitude, and brand awareness in this study), and ultimately enhance the equity of their brands. Although WOM is formally outside the control of the company, Godes et al. (2005) suggested four strategies for them to manage WOM; that is, by playing the role of an observer, moderator, mediator, or participant in WOM communications. As an observer, the company simply gathers information to understand how its customers think about it and what its rivals are doing. As a moderator, the company goes beyond listening to encourage discussion. The company normally realizes its moderator role by setting up a platform to allow customers to exchange information or embrace a customer recommendation program. As a mediator, the company takes control of WOM messages and

disseminates them by itself. In this sense, the company tries to control communication content and channels. Finally, the company could be more active and play the role of a direct participant in WOM communications. This can be achieved via social networks such as Facebook.

Third, managers should pay particular attention to building brand attachment and attitude, as this can greatly enhance brand equity, as well as mediating the relationship between brand image and brand equity. Indeed, for managers and marketers, enhancing the level of attitude and attachment is critical to increasing the effect of brand image on brand equity. Firms should also continue to elevate positive evaluations of their brands, while at the same time safeguarding against negative WOM.

5.4 Limitations and future research directions

As with any other research, this study has limitations. First, our study is cross-sectional in nature; hence, it only deals with identifying relationships, rather than causes. Because brand image normally requires an extended period to influence consumers, it is possible that time may have altered our findings, especially at later stages of the product lifecycle. Therefore, our findings should be interpreted with caution. Future studies could address the relationship between brand image and brand equity based on a longitudinal study for better understanding. Second, among drivers of brand equity presented in the literature, we examined only three variables. Thus, future research could address other brand assets, such as brand loyalty and customer perceptions of brand ethics. Third, this study mainly concentrated on specific moderating variables between brand image and brand equity. Future research should extend the model by examining other possible moderators (e.g., product radicalness, conspicuous consumption, self-congruence, consumer interface quality) at high and low levels. Fourth, in the present study, the participants were students who occupied one homogeneous group with similar thinking patterns. Future research could extend the study to other consumer segments and compare the results with the findings of our research. Fifth, the research was conducted in Malaysia. Since different cultures may involve different attitudes, it would be sensible to duplicate the present framework in other countries and note the similarity (or dissimilarity) of findings relative to our study. Sixth, the present research employed only four search goods (i.e., clothing, shoes, handbags, and wallets) and four experience goods (i.e., deodorant, shampoo, clocks, and watches). Future research could replicate the study in other industries to gain more insight. Finally, the setting for marketers and brands may have changed dramatically due to the influence of digitalization. In particular, the understanding of WOM has shifted against the background of social networks. Therefore, future studies should consider social media and new technologies in WOM, as this could make significant contributions to the field. In conclusion, this research appears to be one of the few empirical studies investigating brand equity that conceptualizes direct and contingent linkages in a single analysis. Given the paramount importance of branding in today's competitive setting, more research is needed to enable us to understand how brand equity can further be enhanced.

References

- Aaker DA (1991) Managing brand equity. The Free Press, New York NY
- Aaker DA (1996) Measuring brand equity across products and markets. *Calif Manag Rev* 38(3):102–120
- Ailawadi KL, Lehmann DR, Neslin SA (2003) Revenue premium as an outcome measure of brand equity. *J Mark* 67:1–17
- Akbar US, Azhar SM (2011) The drivers of brand equity : brand image, brand satisfaction and brand trust. Sukkur Institute of Business Administration, Sukkur, pp 1–18
- Alam SS, Yasin NM (2010) The antecedents of online brand trust: Malaysian evidence. *J Bus Econ Manag* 11(2):210–226
- Alba J, Chattopadhyay A (1986) Salience effects in brand recall. *J Mark Res* 23(4):363–369
- Alba JW, Hutchinson JW (1987) Dimensions of consumer expertise. *J Consum Res* 13(4):411–454
- Allsop DT, Bassett BR, Hoskins JA (2007) Word of mouth research: principles and applications. *J Adv Res* 47(4):398–411
- Ambler T, Bhattacharya CB, Edell J, Keller KL, Lemon KN, Mittal B (2002) Relating brand and customer perspectives on marketing management. *J Serv Res* 5(1):13–25
- Anwar Mir I (2013) Examination of attitudinal and intentional drivers of non-deceptive counterfeiting in a south Asian context. *J Bus Econ Manag* 14(3):601–615
- Arndt J (1967) Rollout product-related conversations in the diffusion of a new product. *J Mark Res* 4:291–295
- Bailey R, Ball S (2006) An exploration of the meanings of hotel brand equity. *Serv Ind J* 26(1):15–38
- Baker W, Hutchinson JW, Moore D, Nedungad P (1986) Brand familiarity and advertising: effects on the evoked set and brand preference. *Adv Consum Res* 13(1):637–642
- Bambauer-Sachse S, Mangold S (2011) Brand equity dilution through negative online word-of-mouth communication. *J Retail Consum Serv* 18(1):38–45
- Baron RM, Kenny DA (1986) The moderator-mediator variable distinction in social psychological research: conceptual, strategic, and statistical considerations. *J Pers Soc Psychol* 51(6):1173–1182
- Bei LT, Widdows R (2004) Consumers' online information search behavior and the phenomenon of search vs. experience products. *J Fam Econ Issues* 25(4):449–467
- Belaid S, Behi AT (2011) The role of attachment in building consumer-brand relationships: an empirical investigation in the utilitarian consumption context. *J Prod Brand Manag* 20(1):37–47
- Bentler PM, Chou CP (1987) Practical issues in structural modeling. *Sociol Methods Res* 16(1):78–117
- Berry LL, Parasuraman A, Zeithaml VA (1994) Improving service quality in America: lessons learned. *Acad Manag Exec* 8(2):32–45
- Bettman JR, Park CW (1980) Effects of prior knowledge and experience and phase of the choice process on consumer. *J Consum Res* 7:234–248
- Bone PF (1992) Determinants of WOM communication during product consumption. *Adv Consum Res* 19:579–583
- Brockman BK, Morgan RM (2003) The role of existing knowledge in new product innovativeness and performance. *Decis Sci* 34(2):385–419
- Broniarczyk SM, Alba JW (1994) The importance of the brand in brand extension. *J Mark Res* 31:214–228
- Brown JJ, Reingen PH (1987) Social ties and word-of-mouth referral behavior. *J Consum Res* 14(3):350–362
- Brown J, Broderick AJ, Lee N (2007) Word of mouth communication within online communities: conceptualizing the online social network. *J Interact Mark* 21(3):2–20
- Bryman A, Cramer D (1999) Qualitative data analysis with SPSS release 8 for windows: a guide for social sciences. Routledge, London
- Buil I, Chernatony LD, Hem LE (2009) Brand extension strategies: perceived fit, brand type, and culture influences. *Eur J Mark* 43(11/12):1300–1324
- Buil I, Chernatony LD, Martínez E (2013) Examining the role of advertising and sales promotions in brand equity creation. *J Bus Res* 66(1):115–122
- Chahal H, Bala M (2012) Significant components of service brand equity in healthcare sector. *Int J Health Care Qual Assur* 25(4):343–362
- Charumbira LT, Stotlar DK (2015) Unpacking the integrative conceptual Framework for assessing perceived brand equity in professional sports teams research journals. *J Sports Manag* 1(2):1–15

- Chattopadhyay A, Alba JW (1988) The situational in importance of recall and inference consumer decision making. *J Consum Res* 15(1):1–12
- Chen YS (2010) The drivers of green brand equity: green brand image, green satisfaction, and green trust. *J Bus Ethics* 93(2):307–319
- Chevalier JA, Mayzlin D (2006) The effect of word of mouth on sales: online book reviews. *J Mark Res* 43(3):345–354
- Chiang K-P, Dholakia RR (2003) Factors driving consumer intention to shop online: an empirical investigation. *J Consum Psychol* 13(1/2):177–183
- Christodoulides G, Chernatony Ld (2010) Consumer based brand equity conceptualization and measurement: a literature review. *Int J Res Mark* 52(1):43–66
- Cobb-Walgren CJ, Ruble CA, Donthu N (1995) Brand equity, brand preference, and purchase intent. *J Adv* 24(3):25–40
- Collier DA (1995) Modelling the relationships between process quality errors and overall service process performance. *Int J Serv Ind Manag* 6(4):4–19
- Cordell VV (1992) Effects of consumer preferences for foreign sourced products. *J Int Bus Stud* 23(2):251–269
- Dabholkar PA, Bagozzi RP (2002) An attitudinal model of technology-based self-service: moderating effects of consumer traits and situational factors. *J Acad Mark Sci* 30(3):184–201
- Djakeli K (2012) Matrix of brand awareness and a positive image as a success factor in political PR. *Sci J Humanit* 1(1):31–35
- Duana W, Gub B, Whinston AB (2008) The dynamics of online word-of-mouth and product sales—an empirical investigation of the movie industry. *J Retail* 84(2):233–242
- Eastman JK, Eastman KL (2011) Perceptions of status consumption and the economy. *J Bus Econ Res* 9(7):9–19
- Emari H, Jafari A, Mogaddam M (2012) The mediatory impact of brand loyalty and brand image on brand equity. *Afr J Bus Manage* 6(17):5692–5701
- Engel JF, Blackwel RD, Kegerreis RJ (1969) How information is used to adopt an innovation. *J Adv Res* 9:3–8
- Esch F, Langner T, Schmitt BH, Geus P (2006) Are brands forever? How brand knowledge and relationships affect current and future purchases. *J Prod Brand Manag* 15(2):98–105
- Faircloth JB, Capella LM, Alford BL (2001) The effect of brand attitude and brand image on brand equity. *J Mark Theory Pract* 9(3):61–75
- Fang YH (2012) Does online interactivity matter? Exploring the role of interactivity strategies in consumer decision making. *Comput Hum Behav* 28(5):1790–1804
- Fatema M, Azad MAK, Masum AKM (2013) Impact of brand image and brand loyalty in measuring. *Asian Bus Rev* 2(1):42–46
- Fedorikhin A, Park CW, Thomson M (2008) Beyond fit and attitude: the effect of emotional attachment on consumer responses to brand extensions. *J Consum Psychol* 18(4):281–291
- Feldman SP, Spencer MC (1965) The effect of personal influence in the selection of consumer services. In: *Proceedings of the fall conference of the american marketing association*. Chicago: American Marketing Association, pp 440–452
- Fornell C, Larcker DF (1981) Evaluating structural equation models with unobservable variables and measurements error. *J Mark Res* 18(1):39–50
- Frazier PA, Barron KE, Tix AP (2004) Testing moderator and mediator effects in counseling psychology research. *J Couns Psychol* 51(1):115–134
- Fullerton G (2005) The impact of brand commitment on loyalty to retail service brands. *Can J Adm Sci* 22(2):97–110
- Gardner BB, Levy SJ (1955) The product and the brand. *Harvard Bus Rev* 33(2):33–39
- Gill MS, Dawra J (2010) Evaluating Aaker's sources of brand equity and the mediating role of brand image. *J Target Measurement Anal Mark* 18(3/4):189–198
- Godes D, Mayzlin D (2004) Using online conversations to study word-of-mouth communication. *Mark Sci* 23(4):545–560
- Godes D, Mayzlin D, Chen Y, Das S, Dellarocas C, Pfeiffer B, Libai B et al (2005) The firm's management of social interactions. *Mark Lett* 16(3/4):415–428
- Goyette I, Ricard L, Bergeron J, Marticotte F (2010) e-WOM scale: word-of-mouth measurement scale for e-services context. *Can J Adm Sci* 27(1):5–23
- Graeff TR (1997) Consumption situations and the effects of brand image on consumers' brand evaluations. *Psychol Mark* 14(1):49–70

- Hair JF, Black WC, Babin BJ, Anderson RE (2010) *Multivariate data analysis*, 7th edn. Prentice Hall, Englewood Cliffs
- Hamzaoui-Essoussi L, Merunka D, Bartikowski B (2011) Brand origin and country of manufacture influences on brand equity and the moderating role of brand typicality. *J Bus Res* 64(9):973–978
- He H, Li Y (2011) Key service drivers for high-tech service brand equity: the mediating role of overall service quality and perceived value. *J Mark Manag* 27(1/2):77–99
- Herr PM, Karde FR, Kim J (1991) Effects of word-of-mouth and product-attribute information on persuasion: an accessibility-diagnostics perspective. *J Consum Res* 17(4):454–462
- Hoch SJ, Ha YW (1986) Consumer learning: advertising and the ambiguity of product experience. *Journal of Consumer Research* 13(2):221–233
- Holden SJS (1992) Brand equity through brand awareness: measuring and managing brand retrieval. Ph.D. Dissertation: University of Florida
- Hsieh MH, Pan SL, Setiono R (2004) Product, corporate, and country image dimensions and purchase behavior: a multicountry analysis. *J Acad Mark Sci* 32(3):251–270
- Huang R, Sarigöllü E (2012) How brand awareness relates to market outcome, brand equity, and the marketing mix. *J Bus Res* 65(1):92–99
- Huang YT, Tsai YT (2013) Antecedents and consequences of brand-oriented companies. *Eur J Mark* 47(11/12):2020–2041
- Huang P, Lurie NH, Mitra S (2009) Searching for experience on the web: an empirical examination of consumer behavior for search and experience goods. *J Mark* 73(2):55–69
- Hung C (2008) The Effect of brand image on public relations perceptions and customer loyalty. *Int J Manag* 25(2):237–247
- Im I, Kim Y, Han HJ (2008) The effects of perceived risk and technology type on users' acceptance of technologies. *Inf Manag* 45(1):1–9
- Ismail AR, Spinelli G (2012) Effects of brand love, personality and image on word of mouth. *J Fashion Mark Manag* 16(4):386–398
- Jacoby J, Szybillo GJ, Busato-Schach J (1977) Information acquisition behavior in brand choice situations. *J Consum Res* 3(4):209–216
- Jaiprakash AT (2008) A conceptual research on the association between celebrity endorsement, brand image and brand equity. *Icfai Univ J Mark Manag* 7(4):54–64
- Jalilvand MR, Samiei N (2012) The effect of electronic word of mouth on brand image and purchase intention: an empirical study in the automobile industry in Iran. *Mark Intell Plann* 30(4):460–476
- Kakati RP, Choudhury S (2013) Measuring customer-based brand equity through brand building blocks for durables. *J Brand Manag* 10(2):24–41
- Kamdar D, Dyne LV (2007) The joint effects of personality and workplace social exchange relationships in predicting task performance and citizenship performance. *J Appl Psychol* 92(5):1286–1298
- Kay MJ (2006) Strong brands and corporate corporate. *Eur J Mark* 40(7/8):742–760
- Kayaman R, Arasli H (2007) Customer based brand equity: evidence from the hotel industry. *Manag Serv Qual* 17(1):92–109
- Keller KL (1993) Conceptualizing, measuring, and managing customer-based brand equity. *J Mark* 57:1–22
- Keller KL (2001) Building customer-based brand equity: a blueprint for creating strong brands. Marketing Science Institute Working paper, 1–107
- Keller E (2007) Unleashing the power of word of mouth: creating brand advocacy to drive growth. *J Adv Res* 47(4):448–452
- Keller KL, Lehmann DR (2003) How do brands create value? *Mark Manag* 12:26–31
- Kim HS (2000) Examination of brand personality and brand attitude within the apparel product category. *J Fash Mark Manag* 4(3):243–252
- Kim H, Kim W, An A (2003) The effect of consumer-based brand equity on firms' financial performance. *J Consum Mark* 20(4):335–351
- Kim KH, Kim KS, Kim DY, Kim JH, Kang SH (2008) Brand equity in hospital marketing. *J Bus Res* 61(1):75–82
- King MF, Balasubramanian SK (1994) The effects of expertise, end goal, and product type on adoption of preference formation strategy. *J Acad Mark Sci* 22(2):146–159
- Klein LR (1998) Evaluating the potential of interactive media through a new lens: search versus experience goods. *J Bus Res* 41(3):195–203
- Kreuzbauer R, Malter AJ (2005) Embodied cognition and new product design: changing product form to influence brand categorization. *J Prod Innov Manag* 22(2):165–176

- Krishnan H (1996) Characteristics of memory associations: a consumer-based brand equity perspective. *Int J Res Mark* 13:389–405
- Kukar-Kinney M, Ridgway NM, Monroe KB (2012) The role of price in the behavior and purchase decisions of compulsive buyers. *J Retail* 88(1):63–71
- Kumaravel V, Kandasamy C (2012) To what extent the brand image influence consumers' purchase decision on durable products. *Rom J Mark* 1:34–38
- Lam D, Lee A, Mizerski R (2009) The effects of cultural values in word-of-mouth communication. *J Int Mark* 17(3):55–70
- Lane V, Jacobson R (1995) Stock market reactions to brand extension announcements: the effects of brand attitude and familiarity. *J Mark* 59(1):63–77
- Laroche M, Kim C, Zhou L (1996) Brand familiarity and confidence as determinants of purchase intention: an empirical test in a multiple brand context. *J Bus Res* 37(2):115–120
- Lassar W, Mittal B, Sharma A (1995) Measuring customer-based brand equity. *J Consum Mark* 12(4):11–19
- Liao S, Cheng CC (2014) Brand equity and the exacerbating factors of product innovation failure evaluations: a communication effect perspective. *J Bus Res* 67(1):2919–2925
- Liao Y-K, Wu W-Y, Amaya Rivas AA, Ju TL (2017) Cognitive, experiential, and marketing factors mediate the effect of brand personality on brand equity. *Social Behav Personal* 45(1):1–18
- Lim BC, Chung CM (2011) The impact of word-of-mouth communication on attribute evaluation. *J Bus Res* 64(1):18–23
- Loken B, John DR (1993) Diluting brand beliefs: when do brand extensions have a negative impact? *J Mark* 57:71–84
- Louis D, Lombart C (2010) Impact of brand personality on three major relational consequences (trust, attachment, and commitment to the brand). *J Prod Brand Manag* 19(2):114–130
- Lovett MJ, Peres R, Shachar R (2013) On brands and word of mouth. *J Mark Res* 50(4):427–444
- MacCallum RC, Zhang S, Preacher KJ, Rucker AD (2002) On the practice of dichotomization of quantitative variables. *Psychol Methods* 7(1):19–40
- Macdonald EK, Sharp BM (2000) Brand awareness effects on consumer decision making for a common, repeat purchase product: a replication. *J Bus Res* 48(1):5–15
- Malär L, Krohmer H, Hoyer WD, Nyffenegger B (2011) Emotional brand attachment and brand personality: the relative importance of the actual and the ideal self. *J Mark* 75(4):35–52
- Martin GS, Brown TJ (1990) In search of brand equity: the conceptualization and measurement of the brand impression construct. *Mark Theory Appl* 2:431–438
- Meyer BO (2012) Emotional and cognitive incorporation of brand equity in financial markets: a conceptualisation of brand equity and behavioural finance. Working Paper, 1–19
- Misner IR (1999) The world's best known marketing secret: building your business with word-of-mouth marketing, 2nd edn. Bard Press, Austin
- Molinari LK, Abratt R, Dion P (2008) Satisfaction, quality and value and effects on repurchase and positive word-of-mouth behavioral intentions in a B2B services context. *J Serv Mark* 22(5):363–373
- Morrin M (1999) The impact of brand extensions on parent brand memory structures and retrieval processes. *J Mark Res* 36:517–525
- Mourad M, Ennew C, Kortam W (2011) Brand equity in higher education. *Mark Intell Plann* 29(4):403–420
- Nedungadi P (1990) Recall and consumer consideration sets: influencing choice without altering brand evaluations. *J Consum Res* 17(3):263–276
- Nelson P (1974) Advertising as information. *J Polit Econ* 82(4):729–754
- Ng PF, Butt MM, Khong KW, Ong FS (2014) Antecedents of green brand equity: an integrated approach. *J Bus Ethics* 121(2):203–215
- Nunnally JC (1978) Psychometric theory. McGraw-Hill, New York, NY
- Oh H (2000) The effect of brand class, brand awareness, and price on customer value and behavioral intentions. *J Hosp Tour Res* 24(2):136–162
- Padilla J, Watson N (2010) A critical review of the literature on generic packaging for cigarettes. A report for PMI. LECG Consulting Belgium, Brussels
- Park CW, Jaworski BJ, MacInnis DJ (1986) Strategic brand concept-image management. *J Mark* 50:135–145
- Park C, MacInnis DJ, Priester J, Eisingerich AB, Iacobucci D (2010) Brand attachment and brand attitude strength: conceptual and empirical differentiation of two critical brand equity drivers. *J Mark* 74:1–17

- Pitta DA, Katsanis LP (1995) Understanding brand equity for successful brand extension. *J Consum Mark* 12(4):51–64
- Plummer JT (2007) Editorial: word of mouth: New advertising discipline? *J Adv Res* 47(4):385–386
- Porter SS, Claycomb C (1997) The influence of brand recognition on retail store image. *J Prod Brand Manag* 6(6):373–387
- Priester JR, Nayakankuppam D, Fleming MA, Godek J (2004) Model: the influence of attitudes and attitude strength on consideration and choice. *J Consum Res* 30(4):574–587
- Roy D, Banerjee S (2007) CARE-ing strategy for integration of brand identity with brand image. *Int J Commer Manag* 17(1):140–148
- Sasmitha J, Mohd Suki N (2015) Young consumers' insights on brand equity. *Int J Retail Distrib Manag* 43(3):276–292
- Sellers-Rubio R, Más-Ruiz FJ (2015) Economic efficiency of members of protected designations of origin: sharing reputation indicators in the experience goods of wine and cheese. *RMS* 9(1):175–196
- Senecal S, Nantel J (2004) The influence of online product recommendations on consumers' online choices. *J Retail* 80(2):159–169
- Seno D, Lukas BA (2007) The equity effect of product endorsement by celebrities: a conceptual framework from a co-branding perspective. *Eur J Mark* 41(1/2):121–134
- Shabbir J, Rehman KU (2013) Impact of perceptual dimensions and behavioral dimension on brand equity in Pakistan. *Inf Manag Bus Rev* 5(7):347–359
- Sheth JN (1971) Word-of-mouth in low-risk innovations. *J Adv Res* 11(3):15–18
- Smith G (2004) Brand image transfer through sponsorship: a consumer learning perspective. *J Mark Manag* 20(3–4):457–474
- Spears N, Singh SN (2004) Measuring attitude toward the brand and purchase intentions. *J Current Issues Res Adv* 26(2):53–66
- Srikatanyoo N, Gnoth J (2002) Country image and international tertiary education. *J Brand Manag* 10(2):139–146
- Srivastava RK (2009) Measuring brand strategy—can brand equity and brand score be a tool to measure the effectiveness of strategy? *Innov Mark* 5(1):26–32
- Strull TK, Wyer RS (1989) Person memory and judgment. *Psychol Rev* 96(1):58–83
- Stigler GJ (1961) The Economics of Information. *J Polit Econ* 69(3):213–225
- Subramaniam A, Al Mamun A, Yukthamarani Permarupan P, Raihani Binti Zainol N (2014) Effects of brand loyalty, image and quality on brand equity: a study among Bank Islam consumers in Kelantan, Malaysia. *Asian Soc Sci* 10(14):67–73
- Thomson M (2006) Human brands: investigating antecedents to consumers' strong attachments to celebrities. *J Mark* 70(3):104–119
- Thomson M, MacInnis DJ, Park CW (2005) The ties that bind: measuring the strength of consumers' emotional attachments to brands. *J Consum Psychol* 15(1):77–91
- Thorelli HB, Lim JS, Ye J (1989) Relative importance of country of origin, warranty, and retail store image on product evaluations. *Int Mark Rev* 6(1):35–46
- Tong C, Wong A, Leung S (2013) The mediating effects of service charge transparency on the relationship between corporate social responsibility and customer behaviour in Hong Kong's retail banking sector. *Bus Econ Res* 3(1):56–88
- Trusov M, Bucklin RE, Pauwels K (2009) Effects of word-of-mouth versus traditional marketing: findings from an internet social networking site. *J Mark* 73:90–102
- Vázquez-Casielles R, Suárez-Álvarez L, Del Río-Lanza AB (2013) The word of mouth dynamic: how positive (and negative) WOM drives purchase probability: An analysis of interpersonal and non-interpersonal factors. *J Adv Res* 53(1):43–60
- Verhoef PC, Franses PH, Hoekstra JC (2002) The effect of relational constructs on customer referrals and number of services purchased from a multiservice provider: does age of relationship matter? *J Acad Mark Sci* 30(3):202–216
- Vieceli J, Shaw RN (2010) Brand salience for fast-moving consumer goods: an empirically based model. *J Mark Manag* 26(13/14):1218–1238
- Voester J, Ivens B, Leischnig A (2016) Partitioned pricing: review of the literature and directions for further research. *Rev Manag Sci*. doi:[10.1007/s11846-016-0208-x](https://doi.org/10.1007/s11846-016-0208-x)
- Wang Y, Kandampully JA, Lo HP, Shi G (2006) The roles of brand equity and corporate reputation in CRM: a Chinese study. *Corp Reput Rev* 9(3):179–197
- Wangenheim Fv (2003) Opportunism in interpersonal exchange: when dissatisfaction is followed by positive word-of-mouth. In: American marketing association summer conference proceedings

- Weiss AM, Anderson E, MacInnis DJ (1999) Reputation management as a motivation for sales structure decisions. *J Mark* 63:74–89
- Wilcox JB, Laverie DA, Kolyesnikova N, Duhan DF, Dodd TH (2008) Facets of brand equity and brand survival: a longitudinal examination. *Int J Wine Bus Res* 20(3):202–214
- Wright AA, Lynch JG (1995) Communication effects of advertising versus direct experience when both search and experience attributes are present. *J Consum Res* 21(4):708–718
- Xia L, Bechwati NN (2010) Word of mouse: the role of cognitive personalization in online consumer reviews. *J Interact Adv* 9(1):3–13
- Xu JB, Chan A (2010) A conceptual framework of hotel experience and customer-based brand equity. *Int J Contemp Hosp Manag* 22(2):174–193
- Yasin B (2009) The role of gender on Turkish consumers' decision-making styles, advances in consumer research. *Asia-Pac Conf Proc* 8:301–308
- Yasin NM, Noor MN, Mohamad O (2007) Does image of country-of-origin matter to brand equity? *J Prod Brand Manag* 16(1):38–48
- Yoo B, Donthu N, Lee S (2000) An examination of selected marketing mix elements and brand equity. *J Acad Mark Sci* 28(2):195–211
- Zeithaml VA (2000) Service quality, profitability, and the economic worth of customers: what we know and what we need to learn. *J Acad Mark Sci* 28(1):67–85
- Zhang H, Ko E, Lee E (2013) Moderating effects of nationality and product category on the relationship between innovation and customer equity in Korea and China. *J Prod Innov Manag* 30(1):110–122